

PRODUCT REQUIREMENTS & TECHNICAL SPECIFICATION: NEXUS API v3.0

Document Status: Approved for Development

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Target Release: Q4 2026

1. EXECUTIVE SUMMARY

Nexus API v3.0 is a high-throughput multimodal ingestion pipeline designed to process high-resolution enterprise documents, OCR streams, and real-time structured data extraction with sub-500ms latency.

2. CORE SYSTEM ARCHITECTURE & REQUIREMENTS

- Requirement 1 (R1): Must support asynchronous batch processing of up to 10,000 PDF pages per minute.
- Requirement 2 (R2): Authentication must utilize OAuth2 + JWT tokens with RS256 signing and 1-hour expiration.
- Requirement 3 (R3): All customer documents at rest must be encrypted with AES-256-GCM.
- Requirement 4 (R4): 99.99% service level agreement (SLA) uptime across multi-region deployment.

3. STAKEHOLDER RESPONSIBILITIES

- Backend Team: Implement FastAPI microservices and vector indexing by September 10, 2026.
- Security Audit Team: Perform penetration testing and ISO 27001 compliance audit by September 25, 2026.
- QA Team: Run load tests simulating 50,000 concurrent requests by October 5, 2026.

4. RISK & VULNERABILITY ANALYSIS

- Critical Risk (C1): Third-party OCR parser rate-limiting could throttle peak ingestion volume. Fallback engine required.
- High Risk (H1): Memory leaks in PDF rendering subprocesses under high concurrency. Must implement process isolation.
- Compliance Risk (M1): GDPR right-to-be-forgotten requires automated hard-deletion across all database replicas within 72 hours.

5. KEY MILESTONES & SPRINT TIMELINE

- Sprint 1 Architecture Freeze: August 30, 2026
- Sprint 2 Alpha Internal Release: September 18, 2026
- Security Audit & Pen-Test Completion: September 30, 2026
- Production Go-Live Launch: October 15, 2026

6. APPROVAL & SIGN-OFF

Chief Technology Officer: Signed (David Chen) - August 18, 2026

Head of Security: Signed (Rachel Adams) - August 19, 2026